

Generation Interconnection System Impact Study Report

For Queue Project Q15-0478

Roanoke Rapids 345 kV

December 20, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0478. The Interconnection Customer proposes a 353.3 MW natural gas combined cycle facility interconnecting at the Roanoke Rapids 345 kV substation in the North Carolina area. The request was received on May 15, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 228 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,628,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DEEN PT - GOLDFIELD 138 KV	MCCARTNEY - N TOWER TAP 345 KV	250.3	214	7.4
TYLER - KEWAUNEE 138 KV	MCCARTNEY - NORTH TOWER TAP 345 KV	225.4	227	7.4
DELTAPORT - MOONSTONE 138 KV L	SKAGGS ISLAND JCT 2 - MITCHELL DAM 345 KV	201.3	222	5.5
CEDAR BAYOU PLANT - CENTRALIA 138 KV	MITCHELL DAM - SKAGGS ISLAND JCT 2 345 KV	199.2	208	5.7
JIM HILL - HAGGIN 138 KV	MITCHELL DAM - SKAGGS ISLAND JUNCTION 2 345 KV	197.8	257	5.7
DELTAPORT - POTOSI 138 KV	SKAGGS ISLAND JUNCTION 2 - MITCHELL DAM 345 KV	195.0	246	5.5
JIM HILL - CEDAR BAYOU PLANT 138 KV L	MITCHELL DAM - SKAGGS ISLAND JCT 2 345 KV	193.1	236	5.7
POTOSI - TYLER 138 KV	MITCHELL DAM - SKAGGS ISLAND JCT 2 345 KV	188.3	269	5.5
HARVELL - OYSTER RIV 138 KV	NORTH TOWER TAP - HAGGIN 345 KV	172.7	191	5.4
KERR DAM - TECUMSEH ENERGY CENTER 345 KV	MCCARTNEY - N TOWER TAP 345 KV Line	168.8	1,116	47.5
N TOWER TAP SUBSTATION - CEDAR FALLS WI 138 KV	N TOWER TAP - HAGGIN 345 KV	168.3	304	7.3
HAGGIN - WALTHALL 138 KV	HAGGIN - NORTH TOWER TAP 345 KV line	165.5	439	10.4
OYSTER RIV SUBSTATION - CEDAR FALLS WI 138 KV	N TOWER TAP - HAGGIN 345 KV Line	164.9	212	5.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MITCHELL DAM - SYL LASKIN 138 KV	MITCHELL DAM - HAGGIN SUBSTATION 345 KV Line	163.2	228	5.9
SPRING ST - CHIPOLA 138 KV	MITCHELL DAM - SKAGGS ISLAND JCT 2 345 KV L	162.2	234	7.5
KEWAUNEE - GOLDFIELD 138 KV	MCCARTEY - NORTH TOWER TAP 345 KV Line	157.3	341	7.4
MCCARTNEY - ROANOKE RAPIDS 345 KV Line	GASTON - ROANOKE RAPIDS 345 KV	156.6	1,398	79.0
HAGGIN 345/138kv XFMR	MITCHELL DAM - HAGGIN SUBSTATION 345 KV	156.0	335	7.3
HARVELL - HARROWGATE SUBSTATION 138 KV	NORTH TOWER TAP - HAGGIN 345 KV line	155.3	192	5.4
MCCARTNEY - N TOWER TAP 345 KV	KERR DAM - TECUMSEH ENERGY CTR 345 KV	151.4	1,698	47.6
LOCKS - DEEN PT 138 KV	MCCARTNEY - NORTH TOWER TAP 345 KV	150.8	376	7.4
SKAGGS ISLAND JCT 2 - MITCHELL DAM 345 KV	BEAR - MECKLENBURG POWER STA 345 KV Line	149.2	1,191	27.3
HAGGIN - N TOWER TAP 345 KV	KER DAM - TECUMSEH ENERGY CTR 345 KV	149.2	1,490	40.5
NATCHEZ - CHANCELLOR ROW 138 KV	MITCHELL DAM - SKAGGS ISLAND JCT 2 SUBSTATION 345 KV	148.7	213	6.6
CHIPOLA - WESTCOAST CELLUFIBRE 138 KV	MITCHELL DAM - SKAGGS ISLAND JUNCTION 2 345 KV	146.9	257	9.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GOLDFIELD - DEEN PT 138 KV Line	Rebuild line with higher-capacity conductor	\$3,146,000	8% (\$261,000)
TYLER - KEWAUNEE 138 KV	Rebuild line with higher-capacity conductor	\$6,010,000	8% (\$499,000)
DELTAPORT - MOONSTONE 138 KV L	Rebuild line with higher-capacity conductor	\$1,870,000	7% (\$134,000)
CEDAR BAYOU PLANT - CENTRALIA 138 KV	Rebuild line with higher-capacity conductor	\$999,000	7% (\$72,000)
JIM HILL SUBSTATION - HAGGIN SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$953,000	7% (\$69,000)
POTOSI - DELTAPORT 138 KV	Rebuild line with higher-capacity conductor	\$6,102,000	7% (\$437,000)
CEDAR BAYOU PLANT - JIM HILL 138 KV	Rebuild line with higher-capacity conductor	\$4,337,000	7% (\$313,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
POTOSI - TYLER 138 KV	Rebuild line with higher-capacity conductor	\$602,000	7% (\$43,000)
HARVELL - OYSTER RIV 138 KV	Rebuild line with higher-capacity conductor	\$1,525,000	11% (\$172,000)
KERR DAM - TECUMSEH ENERGY CENTER 345 KV line	Rebuild line with higher-capacity conductor	\$28,425,000	8% (\$2,246,000)
NORTH TOWER TAP - CEDAR FALLS WI 138 KV	Rebuild line with higher-capacity conductor	\$9,402,000	15% (\$1,443,000)
WALTHALL - HAGGIN 138 KV Line	Rebuild line with higher-capacity conductor	\$3,589,000	8% (\$294,000)
OYSTE RIV - CEDAR FALLS WI 138 KV	Rebuild line with higher-capacity conductor	\$3,312,000	11% (\$373,000)
MITCHELL DAM - SYL LASKIN 138 KV	Rebuild line with higher-capacity conductor	\$2,102,000	6% (\$122,000)
SPRING ST - CHIPOLA 138 KV	Rebuild line with higher-capacity conductor	\$10,972,000	8% (\$834,000)
KEWAUNEE - GOLDFIELD 138 KV line	Rebuild line with higher-capacity conductor	\$5,275,000	8% (\$438,000)
MCCARTNEY - ROANOKE RAPIDS 345 KV Line	Rebuild line with higher-capacity conductor	\$65,045,000	17% (\$10,991,000)
HAGGIN 345/138kV XFMR	Replace transformer with higher-rated unit	\$20,527,000	9% (\$1,846,000)
HARVELL - HARROWGATE 138 KV	Rebuild line with higher-capacity conductor	\$2,990,000	11% (\$337,000)
MCCARTNEY - N TOWER TAP 345 KV	Rebuild line with higher-capacity conductor	\$100,640,000	9% (\$8,727,000)
LOCKS - DEEN PT 138 KV	Rebuild line with higher-capacity conductor	\$1,028,000	8% (\$85,000)
MITCHELL DAM - SKAGGS ISLAND JCT 2 345 KV L	Rebuild line with higher-capacity conductor	\$62,405,000	6% (\$4,016,000)
HAGGIN - N TOWER TAP 345 KV	Rebuild line with higher-capacity conductor	\$50,954,000	8% (\$4,329,000)
NATCHEZ - CHANCELLOR ROW 138 KV	Rebuild line with higher-capacity conductor	\$4,005,000	7% (\$273,000)
CHIPOLA - WESTCOAST CELLUFIBRE 138 KV	Rebuild line with higher-capacity conductor	\$3,716,000	7% (\$274,000)
	Total allocated to project		\$38,628,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.